Name

Mathematics Stage 5 Core

Standard Deviation

Book 1 Standard Deviation

Version: 261006

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Learning Intention ..... **Error! Bookmark not defined.**

Syllabus Content

MA5-DAT-C-01 compares and analyses datasets using summary statistics and graphical representations

Examine standard deviation as a measure of spread

- Identify standard deviation as a measure of spread
- Calculate the standard deviation of a small dataset using digital tools
- Compare small datasets using standard deviation

Standard Deviation

ⓘ Standard Deviation

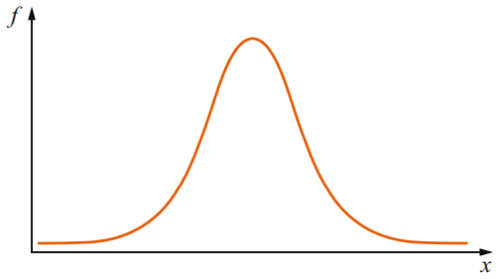
Standard deviation is a number that describes how far data is spread from the **mean**.

It measures the 'average' distance of each data value from the mean using a special formula.

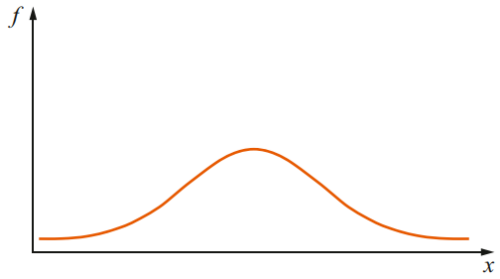
A **larger** standard deviation means **more spread** in the data.

A **lower** standard deviation means **less spread** and **more consistency**.

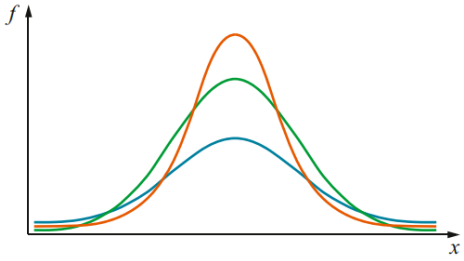
Small standard deviation



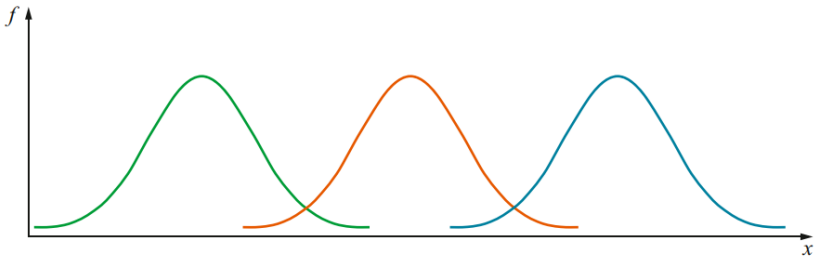
Large standard deviation



Same mean
Different standard deviation



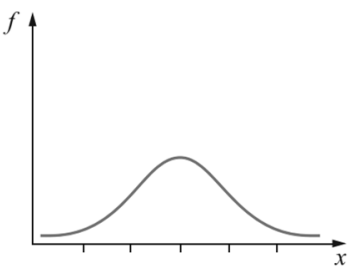
Different mean
Same standard deviation



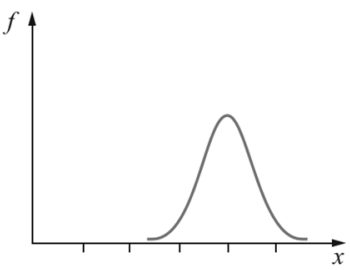
ⓘ **Interpret** Standard deviation

The distribution curves below are drawn with the same scale on each horizontal axis and the same scale on each vertical axis.

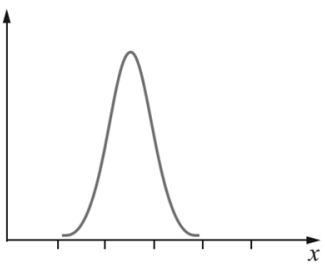
A



B



C



Which distribution has the:

a Largest standard deviation?

b Smallest standard deviation?

c Largest mean?

d Smallest mean?

ⓘ Two Types of Standard Deviation

σ_x **population** standard deviation. (σ is the Greek letter lowercase sigma)
s_x **sample** standard deviation.

Always calculate population standard deviation σ_x,
unless you are specifically asked to calculate sample standard deviation.

ⓘ Standard Deviation using a Calculator

CASIO FX-82 AU

1. [MODE] STAT 1-VAR
2. Enter data using arrow keys and [=]

   x is the outcomes, **Freq** is the probabilities/relative frequencies



   If you do not have a frequency column:

   a. Enter the setup menu [SHIFT] [MODE]

   b. Scroll down [▼] and select STAT

   c. Frequency?
      1:ON 2:OFF

3. Press [AC] to save your data.
4. Enter the data menu [SHIFT] [1] and select 4:Var
5. Calculate standard deviation.



   σ_x [3] population standard deviation.

   s_x [4] sample standard deviation.

   Then press [=]

Standard Deviation using a Calculator

CASIO FX-8200 AU

1. Press HOME
2. Select Statistics
3. Select [1-Variable]
4. Enter data using the arrow keys and EXE
   x is the outcomes, **Freq** is the probabilities/relative frequencies
   If you do not have a frequency column:
   a. Press ○○○ and select [Frequency] > [On]
   b. Press AC to return to the table.
5. After entering your data, press EXE [1-Var Results]
6. Press ∨ for the full list of variables.
   σ_x population standard deviation s_x sample standard deviation.

Example Calculate standard deviation

Calculate population standard deviation σ_x of:

1, 3, 5, 7, 9



Guided Practice

Calculate the population standard deviation, to 2 d.p., of:

a 8, 10, 12

$\sigma_x = 1.63$

b 18, 20, 22

$\sigma_x = 1.63$

c 1, 4, 7, 10, 13

$\sigma_x = 4.24$

d

Score (x)	Frequency (f)
15	3
16	4
17	6
18	4
19	5

$\sigma_x = 1.34$

Key Ideas

1. Standard deviation is a measure of how …… out the data is.
2. Standard deviation is the '……' distance of each datapoint from the ……

1 Calculate the population standard deviation of these datasets. Round to 2 d.p.

a 1, 1, 1, 1, 1

$\sigma_x =$

0

b 1, 1, 2, 3, 3

$\sigma_x =$

0.89

c 2, 2, 3, 4, 4

$\sigma_x =$

0.89

d 4, 4, 6, 8, 8

$\sigma_x =$

1.79

e 3, 4, 6, 8, 9

$\sigma_x =$

2.28

f 2, 4, 6, 8, 10

$\sigma_x =$

2.83

g 1, 2, 3, 4, 5

$\sigma_x =$

1.41

h 0, 1, 2, 3, 4, 5, 6

$\sigma_x =$

2

i 0, 0.5, 1, 1.5, 2, 2.5, 3

$\sigma_x =$

1

j −3, −2.5, −2, −1.5, −1, −0.5, 0

$\sigma_x =$

1

k −3, −2.5, −2, −1, −0.5, 0

$\sigma_x =$

1.08

l 0, 1000

$\sigma_x =$

500

2 The results for Molly and Harry in their tests this term is listed below.

Molly	10	19	16	15	7	15	11	3	9	16
Harry	18	16	17	8	14	15	13	10	13	16

a What is the mean and population standard deviation for Molly's results?

Mean: 12.1, SD: 4.68

b What is the mean and population standard deviation for Harry's results?

Mean: 14, SD: 2.97

c Which student had the more consistent results throughout the term? Give a reason.

Harry — lower standard deviation indicates less spread, so more consistent results.

3 Calculate the population standard deviation from these frequency tables to 2 d.p.

a

x	f
1	1
2	1
3	1

$\sigma_x =$

0.82

b

x	f
1	5
2	5
3	5

$\sigma_x =$

0.82

c

x	f
2	5
3	5
4	5

$\sigma_x =$

0.82

d

x	f
-2	5
-3	5
-4	5

$\sigma_x =$

0.82

e

x	f
-4	5
-6	5
-8	5

$\sigma_x =$

1.63

f

x	f
-4	5
-6	5
-8	50

$\sigma_x =$

1.19

4 During Term 2, Rachel did 9 spelling tests. The tests were scored out of 20.
Her mean score is 15 and her standard deviation is 2.54.

a If Rachel had scored 2 extra marks on every test,

i What would happen to her mean?

Mean increases to 17

ii Explain why her standard deviation would be unchanged.

The spread of the distribution has not changed.

b If Rachel had scored 15 on every test,

i What would happen to her mean?

unchanged

ii Explain why her standard deviation would be zero.

No spread means 0 standard deviation

c What will happen to Rachel's standard deviation if she scores 15 on her next test?

15 is the mean. A value at the mean is 0 distance from the mean, so the standard deviation decreases

d What will happen to Rachel's standard deviation if she scores 0 on her next test?

0 is 15 below the mean, which is much greater than the current SD of 2.54, so adding this value increases the spread and therefore the SD

5 NAPLAN A

In a small business, the seven employees earn the following wages per week:
$300, $490, $520, $590, $660, $680, $970

a Calculate the standard deviation for this set of data.

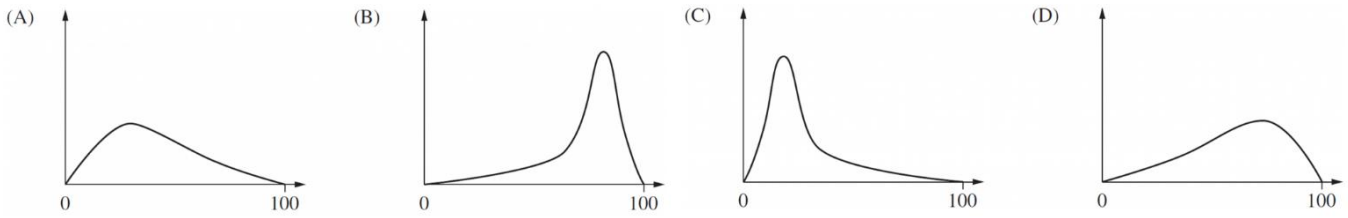
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b Each employee receives a $20 pay increase.
Explain the effect that this increase will this have on the standard deviation.

Standard deviation is unchanged as the amount of spread has not changed

6 2006 HSC Standard 2 Band 5

Which of these graphs best represents positively skewed data with the smaller standard deviation?

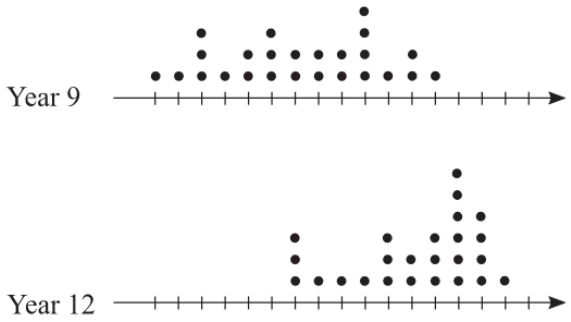


Positively skewed: Tail to the right, so A or C.

Smaller SD: less spread out, so **C**

7 HSC Standard 2 Sample Question Band 5

The dot plots show the height of students in Year 9 and Year 12 in a school.
They are drawn on the same scale.



Which statement about the change in heights when comparing Y9 to Y12 is correct?

A. The mean increased and the standard deviation decreased.

B. The mean decreased and the standard deviation decreased.

C. The mean increased and the standard deviation increased.

D. The mean decreased and the standard deviation increased.

Year 12: clustered more tightly so SD decreased,

centre of data is higher, mean increased. **A**